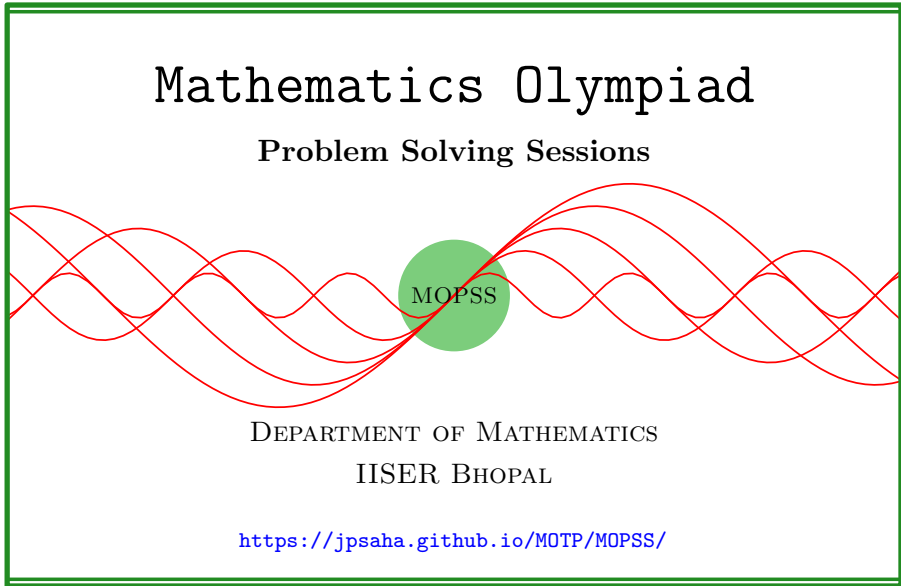


Inequalities

MOPSS



Suggested readings

- Evan Chen's advice [On reading solutions](https://blog.evanchen.cc/2017/03/06/on-reading-solutions/), available at <https://blog.evanchen.cc/2017/03/06/on-reading-solutions/>.
- Evan Chen's [Advice for writing proofs/Remarks on English](https://web.evanchen.cc/handouts/english/english.pdf), available at <https://web.evanchen.cc/handouts/english/english.pdf>.
- [Notes on proofs](#) by Evan Chen from [OTIS Excerpts](#) [[Che25](#), Chapter 1].
- [Tips for writing up solutions](https://www.math.utoronto.ca/barbeau/writingup.pdf) by Edward Barbeau, available at <https://www.math.utoronto.ca/barbeau/writingup.pdf>.
- Evan Chen discusses why [math olympiads](#) are a valuable experience for [high schoolers](#) in the post on [Lessons from math olympiads](#), available at <https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/>.

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§1 Inequalities

§1.1 Warm up

Example 1.1 (Nesbitt's inequality 1903, Moscow MO 1963 Grade 9, UK BMO 1976 P2, India RMO 1990 P2). Let $a, b, c > 0$. Then

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}.$$

This has many proofs, for instance using AM-GM, AM-HM, Cauchy-Schwarz inequality, rearrangement inequality. We present a quick proof from [Hun08].

Solution 1. Put

$$\begin{aligned}\alpha &= \frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b}, \\ \beta &= \frac{b}{b+c} + \frac{c}{c+a} + \frac{a}{a+b}, \\ \gamma &= \frac{c}{b+c} + \frac{a}{c+a} + \frac{b}{a+b}.\end{aligned}$$

By the AM-GM inequality, the inequalities $\alpha + \beta \geq 3$, $\beta + \gamma \geq 3$, $\gamma + \alpha \geq 3$ hold. Adding them together yields $2(\alpha + \beta + \gamma) \geq 9$. Using $\beta + \gamma = 3$, we obtain $\alpha \geq 3/2$. ■

§1.2 No square is negative ($x^2 \geq 0$)

Exercise 1.2 (CMO 1971 P2, AoPS). Prove that if $x > 0, y > 0$ and $x + y = 1$, then

$$\left(1 + \frac{1}{x}\right) \left(1 + \frac{1}{y}\right) \geq 9.$$

Walkthrough —

(a)

Solution 2. Note that

$$\begin{aligned}\left(1 + \frac{1}{x}\right) \left(1 + \frac{1}{y}\right) &= 1 + \frac{1}{x} + \frac{1}{y} + \frac{1}{xy} \\ &= 1 + 1 + \frac{y}{x} + 1 + \frac{x}{y} + \frac{x+y}{xy} \\ &= 1 + 2 \left(2 + \frac{x}{y} + \frac{y}{x}\right) \\ &\geq 9.\end{aligned}$$

■

Exercise 1.3 (USSR Olympiad 1990). Prove that for arbitrary $t \in \mathbb{R}$, the inequality $t^4 - t + \frac{1}{2} > 0$ holds.

Walkthrough —

(a)

Solution 3. For any $t \in \mathbb{R}$, note that

$$\begin{aligned} t^4 - t + \frac{1}{2} &= t^4 - t^2 + \frac{1}{4} + t^2 - t + \frac{1}{4} \\ &= \left(t^2 - \frac{1}{2}\right)^2 + \left(t - \frac{1}{2}\right)^2 \geq 0, \end{aligned}$$

where equality occurs only if $t = \frac{1}{2}$ and $t^2 = \frac{1}{2}$, which is impossible. This shows that $t^4 - t + \frac{1}{2} > 0$ for any real number t . ■

Example 1.4 (India RMO 1995 P7). Show that for any real number x :

$$x^2 \sin x + x \cos x + x^2 + \frac{1}{2} > 0.$$

Solution 4. When $1 + \sin x \neq 0$, we have

$$\begin{aligned} &x^2 \sin x + x \cos x + x^2 + \frac{1}{2} \\ &= (1 + \sin x)x^2 + x \cos x + \frac{1}{2} \\ &= (1 + \sin x) \left(x + \frac{\cos x}{2(1 + \sin x)}\right)^2 + \frac{1}{2} - \frac{\cos^2 x}{4(1 + \sin x)} \\ &= (1 + \sin x) \left(x + \frac{\cos x}{2(1 + \sin x)}\right)^2 + \frac{2 + 2 \sin x - \cos^2 x}{4(1 + \sin x)} \\ &= (1 + \sin x) \left(x + \frac{\cos x}{2(1 + \sin x)}\right)^2 + \frac{(1 + \sin x)^2}{4} > 0. \end{aligned}$$

If $\sin x = -1$, then $x^2 \sin x + x \cos x + x^2 + \frac{1}{2}$ is equal to $\frac{1}{2}$ and hence it is positive. This completes the proof. ■

Example 1.5 (India RMO 2014d P2). If x and y are positive real numbers, prove that

$$4x^4 + 4y^3 + 5x^2 + y + 1 \geq 12xy.$$

Solution 5. Note that

$$4x^4 + 4y^3 + 5x^2 + y + 1 \geq (4x^4 + 1) + 5x^2 + (4y^3 + y)$$

$$\begin{aligned}
 &= (2x^2 - 1)^2 + 9x^2 + y(2y - 1)^2 + 4y^2 \\
 &\geq 9x^2 + 4y^2 \\
 &= (3x - 2y)^2 + 12xy \\
 &\geq 12xy.
 \end{aligned}$$

■

Example 1.6 (India RMO 2014c P2). Find all real x, y such that

$$x^2 + 2y^2 + \frac{1}{2} \leq x(2y + 1).$$

Solution 6. Let x, y be reals satisfying the above inequality. Note that

$$\begin{aligned}
 x^2 + 2y^2 + \frac{1}{2} - x(2y + 1) &= x^2 + 2y^2 + \frac{1}{2} - 2xy - x \\
 &= 2(y^2 - xy) + x^2 - x + \frac{1}{2} \\
 &= 2\left(y - \frac{x}{2}\right)^2 + \frac{x^2}{2} - x + \frac{1}{2} \\
 &= 2\left(y - \frac{x}{2}\right)^2 + \frac{1}{2}(x - 1)^2 \\
 &\geq 0.
 \end{aligned}$$

For real x, y , the given inequality is equivalent to $x - 1 = 0, y - \frac{x}{2} = 0$, which holds if and only if (x, y) is equal to $(1, \frac{1}{2})$. ■

Example 1.7 (India RMO 2015c P7). Let $x, y, z \in \mathbb{R}$, such that $x^2 + y^2 + z^2 - 2xyz = 1$. Prove that

$$(1 + x)(1 + y)(1 + z) \leq 4 + 4xyz.$$

Solution 7. Note that

$$\begin{aligned}
 &4 + 4xyz - (1 + x)(1 + y)(1 + z) \\
 &= 4 + 4xyz - (1 + x + y + z + x^2 + y^2 + z^2 + xyz) \\
 &= 4 + 3xyz - (1 + x + y + z + x^2 + y^2 + z^2) \\
 &= 4 + 3\frac{x^2 + y^2 + z^2 - 1}{2} - (1 + x + y + z + x^2 + y^2 + z^2) \\
 &= \frac{1}{2}(5 + 3(x^2 + y^2 + z^2) - 2(1 + x + y + z + x^2 + y^2 + z^2)) \\
 &= \frac{1}{2}(3 + x^2 + y^2 + z^2 - 2(x + y + z)) \\
 &= \frac{1}{2}((x - 1)^2 + (y - 1)^2 + (z - 1)^2),
 \end{aligned}$$

which is nonnegative. So the required inequality follows. ■

Example 1.8 (Putnam 1998 B1, India RMO 2015f P1). Find the minimum value of

$$\frac{(x + 1/x)^6 - (x^6 + 1/x^6) - 2}{(x + 1/x)^3 + (x^3 + 1/x^3)}$$

for $x \in \mathbb{R}^+$.

Solution 8. Note that

$$\begin{aligned} \frac{(x + 1/x)^6 - (x^6 + 1/x^6) - 2}{(x + 1/x)^3 + (x^3 + 1/x^3)} &= \frac{(x + 1/x)^6 - (x^3 + 1/x^3)^2}{(x + 1/x)^3 + (x^3 + 1/x^3)} \\ &= (x + 1/x)^3 - (x^3 + 1/x^3) \\ &= 3 \left(x + \frac{1}{x} \right) \\ &= 6 + 3 \left(\sqrt{x} - \frac{1}{\sqrt{x}} \right)^2. \end{aligned}$$

It follows that the required minimum value is equal to 6, which is attained at $x = 1$. ■

Example 1.9 (India RMO 2018a P2). Let n be a natural number. Find all real numbers x satisfying the equation

$$\sum_{k=1}^n \frac{kx^k}{1 + x^{2k}} = \frac{n(n+1)}{4}.$$

Solution 9. Let x be a real number satisfying the above equation. It follows that $x \neq 0$. Note that

$$\begin{aligned} \frac{n(n+1)}{4} &= \sum_{k=1}^n \frac{kx^k}{1 + x^{2k}} \\ &= \sum_{k=1}^n \frac{k}{x^k + \frac{1}{x^k}} \\ &\leq \sum_{k=1}^n \frac{k}{|x|^k + \frac{1}{|x|^k}} \\ &\leq \sum_{k=1}^n \frac{k}{2} \\ &= \frac{n(n+1)}{4}. \end{aligned}$$

Consequently, the inequalities in the intermediate steps are equalities. This shows that x is positive and $|x| = 1$, which gives $x = 1$. Also note that the given equation holds if $x = 1$. Hence, $x = 1$ is the only real solution of the given equation. ■

§1.3 Manipulation

Example 1.10 (India INMO 1987 P2). Determine the largest number in the infinite sequence

$$1, 2^{1/2}, 3^{1/3}, 4^{1/4}, \dots, n^{1/n}, \dots$$

Solution 10. Note that $1 < 2^{1/2} < 3^{1/3}$ holds. Let us establish the following Claim.

Claim — For any integer $n \geq 3$, the inequality

$$n^{\frac{1}{n}} > (n+1)^{\frac{1}{n+1}}$$

holds.

Proof of the Claim. Observe that the inequality is equivalent to $n > (1 + \frac{1}{n})^n$. For any integer $n \geq 3$, note that

$$\begin{aligned} & \left(1 + \frac{1}{n}\right)^n \\ &= 1 + \binom{n}{1} \frac{1}{n} + \binom{n}{2} \frac{1}{n^2} + \dots + \binom{n}{n} \frac{1}{n^n} \\ &= 1 + \frac{n}{1!} \frac{1}{n} + \frac{n(n-1)}{2!} \frac{1}{n^2} + \frac{n(n-1)(n-2)}{3!} \frac{1}{n^3} \\ & \quad + \dots + \frac{n(n-1)(n-2) \dots 2}{(n-1)!} \frac{1}{n^{n-1}} + \frac{n(n-1)(n-2) \dots 1}{n!} \frac{1}{n^n} \\ &= 1 + \frac{1}{1!} + \frac{1}{2!} \left(1 - \frac{1}{n}\right) + \frac{1}{3!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) + \\ & \quad \dots + \frac{1}{(n-1)!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{n-2}{n}\right) \\ & \quad + \frac{1}{n!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{n-2}{n}\right) \left(1 - \frac{n-1}{n}\right) \\ &< 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots + \frac{1}{n!} \\ &< 2 + \frac{1}{2} + \frac{1}{2 \cdot 2} + \frac{1}{2 \cdot 2 \cdot 2} + \dots + \underbrace{\frac{1}{2 \cdot 2 \cdot 2 \dots 2}}_{(n-1)\text{-times}} \\ &= 2 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots + \frac{1}{2^{n-1}} \\ &< 3 \\ &\leq n \end{aligned}$$

holds. This proves the Claim. □

It follows that the largest term in the given sequence is equal to $3^{1/3}$. ■

Remark. For any $n \geq e$, note that

$$n^{1/n} \geq e^{1/n} > 1 + \frac{1}{n}$$

holds, which shows that $n^{1/n} > (n+1)^{1/(n+1)}$.

Example 1.11 (India RMO 2000 P3). Suppose $(x_1, x_2, \dots, x_n, \dots)$ is a sequence of positive real numbers such that $x_1 \geq x_2 \geq x_3 \geq \dots x_n \geq \dots$, and for all n ,

$$\frac{x_1}{1} + \frac{x_4}{2} + \frac{x_9}{3} + \dots + \frac{x_{n^2}}{n} \leq 1.$$

Show that for all $k \geq 1$ the following inequality is satisfied:

$$\frac{x_1}{1} + \frac{x_2}{2} + \frac{x_3}{3} + \dots + \frac{x_k}{k} \leq 3.$$

Solution 11. Note that for any integer $n \geq 1$, we have

$$\begin{aligned} & \left(\frac{x_1}{1} + \frac{x_2}{2} + \frac{x_3}{3} \right) \\ & + \left(\frac{x_4}{4} + \frac{x_5}{5} + \frac{x_6}{6} + \frac{x_7}{7} + \frac{x_8}{8} \right) \\ & + \dots + \left(\frac{x_{n^2}}{n^2} + \frac{x_{n^2+1}}{n^2+1} + \dots + \frac{x_{(n+1)^2-1}}{(n+1)^2-1} \right) \\ & \leq \left(\frac{x_1}{1} + \frac{x_1}{1} + \frac{x_1}{1} \right) + \left(\frac{x_4}{4} + \frac{x_4}{4} + \frac{x_4}{4} + \frac{x_4}{4} + \frac{x_4}{4} \right) \\ & + \dots + \underbrace{\left(\frac{x_{n^2}}{n^2} + \frac{x_{n^2}}{n^2} + \dots + \frac{x_{n^2}}{n^2} \right)}_{(2n+1) \text{ terms}} \\ & = (2 \cdot 1 + 1) \frac{x_1}{1} + (2 \cdot 2 + 1) \frac{x_4}{4} + \dots + (2n + 1) \frac{x_{n^2}}{n^2} \\ & \leq (3 \cdot 1) \frac{x_1}{1} + (3 \cdot 2) \frac{x_4}{4} + \dots + (3n) \frac{x_{n^2}}{n^2} \\ & = 3 \left(\frac{x_1}{1} + \frac{x_4}{2} + \dots + \frac{x_{n^2}}{n} \right) \\ & \leq 3. \end{aligned}$$

Since for any $k \geq 1$, there is a positive integer m with $(m+1)^2 - 1 \geq k$, the result follows. ■

Example 1.12 (India RMO 2002 P6). For any natural number $n > 1$, prove the inequality

$$\frac{1}{2} < \frac{1}{n^2+1} + \frac{2}{n^2+2} + \frac{3}{n^2+3} + \dots + \frac{n}{n^2+n} < \frac{1}{2} + \frac{1}{2n}.$$

Solution 12. Note that

$$\begin{aligned} & \frac{1}{n^2+1} + \frac{2}{n^2+2} + \frac{3}{n^2+3} + \cdots + \frac{n}{n^2+n} \\ & > \frac{1}{n^2+n}(1+2+3+\cdots+n) \quad (\text{using } n > 1) \\ & = \frac{1}{2}. \end{aligned}$$

Also note that

$$\begin{aligned} & \frac{1}{n^2+1} + \frac{2}{n^2+2} + \frac{3}{n^2+3} + \cdots + \frac{n}{n^2+n} \\ & < \frac{1}{n^2+1}(1+2+3+\cdots+n) \quad (\text{using } n > 1) \\ & = \frac{n^2+n}{2(n^2+1)} \\ & = \frac{1}{2} + \frac{n-1}{2(n^2+1)} \\ & < \frac{1}{2} + \frac{1}{2n}. \end{aligned}$$

■

Example 1.13 (India RMO 2005 P3). If a, b, c are three real numbers such that $|a-b| \geq |c|$, $|b-c| \geq |a|$, $|c-a| \geq |b|$, then prove that one of a, b, c is the sum of the other two.

Solution 13. The given inequalities are equivalent to $(a-b)^2 - c^2 \geq 0$, $(b-c)^2 - a^2 \geq 0$, $(c-a)^2 - b^2 \geq 0$, which yields

$$\begin{aligned} (a-b+c)(a-b-c) & \geq 0, \\ (b-c+a)(b-c-a) & \geq 0, \\ (c-a+b)(c-a-b) & \geq 0. \end{aligned}$$

Multiplying them, we obtain

$$-(b+c-a)^2(c+a-b)^2(a+b-c)^2 \geq 0,$$

which shows that $(b+c-a)(c+a-b)(a+b-c)$ is equal to 0. This proves the result. ■

§1.4 Rearrangement inequality

Theorem 1 (Rearrangement inequality)

Let $a_1, \dots, a_n, b_1, \dots, b_n$ be real numbers satisfying $a_1 \geq a_2 \geq \dots \geq a_n, b_1 \geq b_2 \geq \dots \geq b_n$. Then for any permutation σ of $\{1, 2, \dots, n\}$,

$$\begin{aligned} a_1 b_1 + a_2 b_2 + \dots + a_n b_n &\geq a_1 b_{\sigma(1)} + a_2 b_{\sigma(2)} + \dots + a_n b_{\sigma(n)} \\ &\geq a_1 b_n + a_2 b_{n-1} + \dots + a_n b_1 \end{aligned}$$

holds. In other words, for any two sequences $a_1, \dots, a_n$ and $b_1, \dots, b_n$ of real numbers, the sum $a_1 b_1 + \dots + a_n b_n$ is maximized (resp. minimized) when these sequences are sorted in the same (resp. opposite) order.

Exercise 1.14 (CMO 2002 P3, AoPS). For positive $x, y, z \in \mathbb{R}$, prove that

$$\frac{x^3}{yz} + \frac{y^3}{zx} + \frac{z^3}{xy} \geq x + y + z.$$

Walkthrough —

(a)

Solution 14. Since x, y, z are positive, it follows that the sequences (x^3, y^3, z^3) and $(\frac{1}{yz}, \frac{1}{zx}, \frac{1}{xy})$ are similarly ordered. By the rearrangement inequality, it follows that

$$\frac{x^3}{yz} + \frac{y^3}{zx} + \frac{z^3}{xy} \geq \frac{x^3}{zx} + \frac{y^3}{xy} + \frac{z^3}{yz} = \frac{x^2}{z} + \frac{y^2}{x} + \frac{z^2}{y}.$$

Since the sequences (x^2, y^2, z^2) and $(\frac{1}{x}, \frac{1}{y}, \frac{1}{z})$ are sorted in the opposite order, using the rearrangement inequality, we get

$$\frac{x^2}{z} + \frac{y^2}{x} + \frac{z^2}{y} \geq \frac{x^2}{x} + \frac{y^2}{y} + \frac{z^2}{z} = x + y + z.$$

Combining the above inequalities, we obtain

$$\frac{x^3}{yz} + \frac{y^3}{zx} + \frac{z^3}{xy} \geq x + y + z.$$

■

Example 1.15. For positive reals a, b, c , show that $a^7 + b^7 + c^7 \geq a^4 b^3 + b^4 c^3 + c^4 a^3$.

Solution 15. Since a, b, c are positive reals, it follows that the sequences a^4, b^4, c^4 and a^3, b^3, c^3 are sorted in the same order. By the rearrangement inequality, we obtain

$$a^7 + b^7 + c^7 \geq a^4b^3 + b^4c^3 + c^4a^3.$$

■

Example 1.16 (India RMO 2012a P3, India RMO 2012b P3, India RMO 2012c P3, India RMO 2012d P3). Let a and b be positive real numbers such that $a + b = 1$. Prove that $a^ab^b + a^bb^a \leq 1$.

Example 1.17 (India RMO 2012a P3, India RMO 2012b P3, India RMO 2012c P3, India RMO 2012d P3). Let a and b be positive real numbers such that $a + b = 1$. Prove that $a^ab^b + a^bb^a \leq 1$.

Example 1.18 (India RMO 2012a P3, India RMO 2012b P3, India RMO 2012c P3, India RMO 2012d P3). Let a and b be positive real numbers such that $a + b = 1$. Prove that $a^ab^b + a^bb^a \leq 1$.

Example 1.19 (India RMO 2012a P3, India RMO 2012b P3, India RMO 2012c P3, India RMO 2012d P3). Let a and b be positive real numbers such that $a + b = 1$. Prove that $a^ab^b + a^bb^a \leq 1$.

Solution 16. For any positive a, b , the sequences $(a^a, b^a), (a^b, b^b)$ are sorted the same way. Applying the rearrangement inequality, we obtain

$$a^ab^b + a^bb^a \leq a^aa^b + b^ab^b = a^{a+b} + b^{a+b} = a + b = 1.$$

■

Example 1.20 (India RMO 2014b P2). Let x, y, z be positive real numbers. Prove that

$$\frac{y^2 + z^2}{x} + \frac{z^2 + x^2}{y} + \frac{x^2 + y^2}{z} \geq 2(x + y + z).$$

Solution 17. Note that

$$\frac{y^2 + z^2}{x} + \frac{z^2 + x^2}{y} + \frac{x^2 + y^2}{z} = \left(\frac{y^2}{x} + \frac{z^2}{y} + \frac{x^2}{z} \right) + \left(\frac{z^2}{x} + \frac{x^2}{y} + \frac{y^2}{z} \right).$$

holds. Since (x^2, y^2, z^2) and $(\frac{1}{x}, \frac{1}{y}, \frac{1}{z})$ are sorted oppositely, by the rearrangement inequality, we obtain

$$\left(\frac{y^2}{x} + \frac{z^2}{y} + \frac{x^2}{z} \right) \geq x + y + z, \quad \left(\frac{z^2}{x} + \frac{x^2}{y} + \frac{y^2}{z} \right) \geq x + y + z.$$

Adding these inequalities, the required inequality follows. ■

Remark. Note that the last two inequalities also follow from the Cauchy–Schwarz inequality.

Example 1.21 (India INMO 2001 P3). If a, b, c are positive real numbers such that $abc = 1$, Prove that

$$a^{b+c}b^{c+a}c^{a+b} \leq 1.$$

Solution 18. Put

$$\begin{aligned}\alpha &= a^{b+c}b^{c+a}c^{a+b}, \\ \beta &= b^{b+c}c^{c+a}a^{a+b}, \\ \gamma &= c^{b+c}a^{c+a}b^{a+b}.\end{aligned}$$

Note that $\alpha\beta\gamma = 1$, and by the rearrangement inequality, it follows that

$$\alpha \leq \beta, \alpha \leq \gamma$$

hold. This gives $\alpha^3 \leq 1$, which yields $\alpha \leq 1$. ■

Remark. The above proof is similar to be the proof of Nesbitt’s inequality as in [Hun08].

Example 1.22 (India RMO 2017a P6). Let x, y, z be real numbers, each greater than 1. Prove that

$$\frac{x+1}{y+1} + \frac{y+1}{z+1} + \frac{z+1}{x+1} \leq \frac{x-1}{y-1} + \frac{y-1}{z-1} + \frac{z-1}{x-1}$$

Solution 19. Note that both sides of the above inequality remain invariant under cyclic permutations of x, y, z . Therefore, without loss of generality, we assume that $x \geq y$ and $x \geq z$. Note that

$$\begin{aligned}\frac{x-1}{y-1} - \frac{x+1}{y+1} &= \frac{(x-1)(y+1) - (x+1)(y-1)}{y^2-1} \\ &= \frac{2(x-y)}{y^2-1}.\end{aligned}$$

Similarly, it follows that

$$\begin{aligned}\frac{y-1}{z-1} - \frac{y+1}{z+1} &= \frac{2(y-z)}{z^2-1}, \\ \frac{z-1}{x-1} - \frac{z+1}{x+1} &= \frac{2(z-x)}{x^2-1}.\end{aligned}$$

It suffices to prove that

$$\frac{x-y}{y^2-1} + \frac{y-z}{z^2-1} + \frac{z-x}{x^2-1} \geq 0,$$

which is equivalent to

$$\frac{x}{y^2-1} + \frac{y}{z^2-1} + \frac{z}{x^2-1} \geq \frac{x}{x^2-1} + \frac{y}{y^2-1} + \frac{z}{z^2-1},$$

which follows from the rearrangement inequality since x, y, z and $\frac{1}{x^2-1}, \frac{1}{y^2-1}, \frac{1}{z^2-1}$ are sorted in the opposite order. ■

§1.5 Cauchy–Schwarz inequality

Exercise 1.23 (All-Russian Mathematical Olympiad 2018 Grade 11 Day 1 P2, AoPS, by F. Petrov). Let $n \geq 2$ and $x_1, x_2, \dots, x_n$ be positive real numbers. Prove that

$$\frac{1+x_1^2}{1+x_1x_2} + \frac{1+x_2^2}{1+x_2x_3} + \dots + \frac{1+x_n^2}{1+x_nx_1} \geq n.$$

Walkthrough —

- (a) Use the Cauchy–Schwarz inequality to obtain a lower bound for each term.
- (b) Use the AM–GM inequality to complete the proof.

Solution 20. Note that

$$(1+a^2)(1+b^2) \geq (1+ab)^2$$

holds for any real numbers a, b by the Cauchy–Schwarz inequality. This implies that

$$\begin{aligned} & \frac{1+x_1^2}{1+x_1x_2} + \frac{1+x_2^2}{1+x_2x_3} + \dots + \frac{1+x_n^2}{1+x_nx_1} \\ & \geq \sqrt{\frac{1+x_1^2}{1+x_2^2}} + \sqrt{\frac{1+x_2^2}{1+x_3^2}} + \dots + \sqrt{\frac{1+x_n^2}{1+x_1^2}} \\ & \geq n, \end{aligned}$$

where the last inequality follows from the AM–GM inequality. This completes the proof. ■

Example 1.24. Show that if the sum of positive numbers a, b, c is equal to 1, then

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 9.$$

Solution 21. Applying the Cauchy–Schwarz inequality, we obtain

$$(a + b + c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \geq 3^2.$$

Using $a + b + c = 1$, the required inequality follows. ■

Example 1.25 (India RMO 2013d P3). Given real numbers $a, b, c, d, e > 1$. Prove that

$$\frac{a^2}{c-1} + \frac{b^2}{d-1} + \frac{c^2}{e-1} + \frac{d^2}{a-1} + \frac{e^2}{b-1} \geq 20.$$

Solution 22. Since $a-1, b-1, c-1, d-1, e-1$ are all positive, by applying the Cauchy–Schwarz inequality, we get

$$\frac{a^2}{c-1} + \frac{b^2}{d-1} + \frac{c^2}{e-1} + \frac{d^2}{a-1} + \frac{e^2}{b-1} \geq \frac{(a+b+c+d+e)^2}{a+b+c+d+e-5}.$$

It suffices to show that

$$\frac{(a+b+c+d+e)^2}{a+b+c+d+e-5} \geq 20$$

holds. Since $a+b+c+d+e-5$ is positive, it is enough to prove that

$$(a+b+c+d+e)^2 - 20(a+b+c+d+e) + 100 \geq 0,$$

which holds

$$(a+b+c+d+e)^2 - 20(a+b+c+d+e) + 100 = (a+b+c+d+e-10)^2.$$

This proves the result. ■

Example 1.26 (India RMO 2014a P6). Let $x_1, x_2, x_3, \dots, x_{2014}$ be positive real numbers such that $\sum_{j=1}^{2014} x_j = 1$. Determine with proof the smallest constant K such that

$$K \sum_{j=1}^{2014} \frac{x_j^2}{1-x_j} \geq 1. \tag{1}$$

Solution 23. For any positive reals $x_1, \dots, x_{2014}$ satisfying $\sum_{j=1}^{2014} x_j = 1$, using the Cauchy–Schwarz inequality, we obtain

$$\sum_{j=1}^{2014} \frac{x_j^2}{1-x_j} \geq \frac{(x_1 + \dots + x_{2014})^2}{2014 - (x_1 + \dots + x_{2014})} = \frac{1}{2013},$$

which gives

$$2013 \sum_{j=1}^{2014} \frac{x_j^2}{1-x_j} \geq 1.$$

This shows that the inequality (1) holds for $K = 2013$.

For $x_1 = x_2 = \dots = x_{2014} = \frac{1}{2014}$, note that

$$\sum_{j=1}^{2014} \frac{x_j^2}{1-x_j} = \frac{1}{2013}$$

holds, which shows that for any K satisfying the inequality (1), the bound $K \geq 2013$ holds.

This proves that the smallest constant K satisfying the inequality (1) is equal to 2013. ■

Example 1.27 (India RMO 2016d P2). Let a, b, c be positive real numbers such that

$$\frac{ab}{1+bc} + \frac{bc}{1+ca} + \frac{ca}{1+ab} = 1.$$

Prove that

$$\frac{1}{a^3} + \frac{1}{b^3} + \frac{1}{c^3} \geq 6\sqrt{2}.$$

We refer to Example 1.60 and Example 1.61.

Solution 24. Applying the Cauchy-Schwarz inequality, we obtain

$$\begin{aligned} 1 &= \frac{ab}{1+bc} + \frac{bc}{1+ca} + \frac{ca}{1+ab} \\ &\geq \frac{(\sqrt{ab} + \sqrt{bc} + \sqrt{ca})^2}{3+ab+bc+ca}, \end{aligned}$$

which yields

$$3 \geq 2\sqrt{abc}(\sqrt{a} + \sqrt{b} + \sqrt{c}).$$

Using the AM-GM inequality, we get

$$2\sqrt{abc} \sqrt[3]{\sqrt{abc}} \leq 1,$$

which gives $abc \leq \frac{1}{2\sqrt{2}}$. Note that

$$\begin{aligned} \frac{1}{a^3} + \frac{1}{b^3} + \frac{1}{c^3} &\geq \frac{3}{abc} \\ &\geq 6\sqrt{2}. \end{aligned}$$

■

Example 1.28 (India RMO 2023b P5). Let $n > k > 1$ be positive integers. Determine all positive real numbers $a_1, a_2, \dots, a_n$ which satisfy

$$\sum_{i=1}^n \sqrt{\frac{ka_i^k}{(k-1)a_i^k + 1}} = \sum_{i=1}^n a_i = n.$$

Solution 25. Let $a_1, \dots, a_n$ be positive reals satisfying the above condition. Note that

$$\begin{aligned}
 & \sum_{i=1}^n \sqrt{\frac{ka_i^k}{(k-1)a_i^k + 1}} \\
 & \leq \sum_{i=1}^n \sqrt{\frac{a_i^k}{\sqrt[k]{a_i^{k(k-1)}}}} \quad (\text{by the AM-GM inequality}) \\
 & = \sum_{i=1}^n \sqrt{a_i} \\
 & \leq \left(n \sum_{i=1}^n a_i \right)^{1/2} \quad (\text{by the Cauchy-Schwarz inequality}) \\
 & = \sum_{i=1}^n \sqrt{\frac{ka_i^k}{(k-1)a_i^k + 1}}
 \end{aligned}$$

hold, and hence, the intermediate inequalities are equalities. This implies that $a_1, a_2, \dots, a_n$ are equal. Using the given condition, it follows that

$$a_1 = a_2 = \dots = a_n = 1.$$

Also note that the given conditions are satisfied if $a_1 = a_2 = \dots = a_n = 1$ holds. This proves that the positive real numbers satisfying the given condition are precisely

$$a_1 = a_2 = \dots = a_n = 1.$$

■

§1.6 QM-AM-GM-HM inequality

Example 1.29. Prove that if $m > 0$, then

$$m + \frac{4}{m^2} \geq 3.$$

Solution 26. Applying the AM-GM inequality, it follows that

$$m + \frac{4}{m^2} = \frac{m}{2} + \frac{m}{2} + \frac{4}{m^2} \geq 3.$$

■

Example 1.30 (India INMO 1988 P4). If $a, b > 0$ with $a + b = 1$, then show that $(a + \frac{1}{a})^2 + (b + \frac{1}{b})^2 \geq \frac{25}{2}$.

Solution 27. Note that

$$\begin{aligned}
 \left(a + \frac{1}{a}\right)^2 + \left(b + \frac{1}{b}\right)^2 &= a^2 + b^2 + \frac{1}{a^2} + \frac{1}{b^2} + 4 \\
 &= a^2 + b^2 + 1 + \frac{b^2}{a^2} + 2\frac{b}{a} + 1 + \frac{a^2}{b^2} + 2\frac{a}{b} + 4 \\
 &= a^2 + b^2 + \left(\frac{a}{b} - \frac{b}{a}\right)^2 + \left(\sqrt{\frac{a}{b}} - \sqrt{\frac{b}{a}}\right)^2 + 12 \\
 &\geq \frac{(a+b)^2}{2} + 12 \\
 &= \frac{25}{2}.
 \end{aligned}$$

■

Example 1.31 (India RMO 1991 P2). If a, b, c and d are any 4 positive real numbers, then prove that

$$\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a} \geq 4.$$

Solution 28. Note that

$$\begin{aligned}
 \frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a} &\geq 2\sqrt{\frac{a}{c}} + 2\sqrt{\frac{c}{a}} \\
 &\geq 4.
 \end{aligned}$$

■

Example 1.32 (All-Russian MO 1991 Grade 11 First Day P3, India RMO 1994 P8). If a, b and c are positive real numbers such that $a + b + c = 1$, prove that

$$(1+a)(1+b)(1+c) \geq 8(1-a)(1-b)(1-c).$$

Solution 29. The given inequality is equivalent to

$$(b+c+2a)(c+a+2b)(a+b+2c) \geq 8(b+c)(c+a)(a+b).$$

Put $x = b + c, y = c + a, z = a + b$. Note that the above inequality can be rewritten as

$$(y+z)(z+x)(x+y) \geq 8xyz,$$

which follows from the AM-GM inequality. ■

Example 1.33 (India RMO 1993 P6). If a, b, c, d are four positive real numbers such that $abcd = 1$, prove that

$$(1+a)(1+b)(1+c)(1+d) \geq 16.$$

Solution 30. It follows by applying the AM-GM inequality to the factors and using $abcd = 1$. ■

Example 1.34. Show that if $a, b, c \geq 0$, then

$$ab(a+b) + bc(b+c) + ca(c+a) \geq 6abc.$$

Example 1.35. [PK74, Problem 62.2] Each of the four numbers a, b, c , and d is positive and less than one. Show that not all four products

$$4a(1-b), 4b(1-c), 4c(1-d), 4d(1-a)$$

are greater than one.

Solution 31. Note that their product is equal to

$$\prod_{x \in \{a, b, c, d\}} 4x(1-x) = \prod_{x \in \{a, b, c, d\}} (1 - (2x-1)^2)$$

which is at most 1. This shows that not all the above four products are greater than one. ■

Example 1.36 (India RMO 2003 P3). Let a, b, c be three positive real numbers such that $a+b+c = 1$. Prove that among the three numbers $a-ab, b-bc, c-ca$, there is one which is at most $\frac{1}{4}$ and there is one which is at least $\frac{2}{9}$.

Solution 32. Note that

$$(a-ab)(b-bc)(c-ca) = a(1-a)b(1-b)c(1-c) \leq \frac{1}{4^3}$$

holds. So among $a-ab, b-bc, c-ca$ there is one which is at most $\frac{1}{4}$.

Also note that

$$\begin{aligned} (a-ab) + (b-bc) + (c-ca) &= 1 - (ab+bc+ca) \\ &\geq 1 - \frac{1}{3}(a+b+c)^2 \\ &\geq \frac{2}{3} \end{aligned}$$

holds. This shows that one of $a-ab, b-bc, c-ca$ is at least $\frac{2}{9}$. ■

Example 1.37 (India RMO 2005 P7). Let a, b, c be three positive real numbers such that $a + b + c = 1$. Let

$$\lambda = \min\{a^3 + a^2bc, b^3 + ab^2c, c^3 + abc^2\}.$$

Prove that the roots of the equation $x^2 + x + 4\lambda = 0$ are real.

Solution 33. We need to show that the discriminant of $x^2 + x + 4\lambda$ is nonnegative, that is, $\lambda \leq \frac{1}{16}$ holds. On the contrary, let us assume that $\lambda > \frac{1}{16}$. This gives

$$a^3 + a^2bc > \frac{1}{16}, b^3 + ab^2c > \frac{1}{16}, c^3 + abc^2 > \frac{1}{16}.$$

Note that

$$a^3 + a^2bc = a^2(a + bc) = a^2(1 - b - c + bc) = a^2(1 - b)(1 - c) > \frac{1}{16}$$

holds. Similarly, it follows that

$$b^2(1 - c)(1 - a) > \frac{1}{16}, c^2(1 - a)(1 - b) > \frac{1}{16}.$$

This implies that

$$(abc(1 - a)(1 - b)(1 - c))^2 > \frac{1}{16^3},$$

which is impossible since

$$a(1 - a) \leq \frac{1}{4}, b(1 - b) \leq \frac{1}{4}, c(1 - c) \leq \frac{1}{4}.$$

Consequently, we obtain $\lambda \leq \frac{1}{16}$, and hence the roots of the equation $x^2 + x + 4\lambda = 0$ are real. ■

Example 1.38. If a, b, c are positive reals, then show that

$$(a^2 + 1)(b^2 + 1)(c^2 + 1) \geq (a + b)(b + c)(c + a).$$

Solution 34. Note that

$$(a^2 + 1)(b^2 + 1) - (a + b)^2 = (ab - 1)^2 \geq 0$$

holds. Similarly, it follows that

$$(b^2 + 1)(c^2 + 1) \geq (b + c)^2, (c^2 + 1)(a^2 + 1) \geq (c + a)^2.$$

Multiplying the above inequalities and using that a, b, c are nonnegative, the required inequality follows. ■

Example 1.39 (India RMO 2006 P3). If a, b, c are three positive real numbers, prove that

$$\frac{a^2 + 1}{b + c} + \frac{b^2 + 1}{c + a} + \frac{c^2 + 1}{a + b} \geq 3.$$

Solution 35. Note that

$$\frac{a^2 + 1}{b + c} + \frac{b^2 + 1}{c + a} + \frac{c^2 + 1}{a + b} \geq \frac{2a}{b + c} + \frac{2b}{c + a} + \frac{2c}{a + b} \geq 3,$$

where the first inequality follows from AM-GM inequality and the second inequality follows from Nesbitt's inequality Example 1.1. ■

Example 1.40 (India RMO 2007 P6). Prove that

- (a) $5 < \sqrt{5} + \sqrt[3]{5} + \sqrt[4]{5}$,
- (b) $8 > \sqrt{8} + \sqrt[3]{8} + \sqrt[4]{8}$,
- (c) $n > \sqrt{n} + \sqrt[3]{n} + \sqrt[4]{n}$ for all integers $n \geq 9$.

Solution 36. Note that

$$\sqrt{5} > 2, \sqrt[3]{5} > \frac{8}{5}, \sqrt[4]{5} > \frac{7}{5}$$

holds, which gives the first inequality. Using $3 > \sqrt{8}, 3 > \sqrt[4]{8}$, we obtain the second inequality. Note that

$$\sqrt{n} + \sqrt[3]{n} + \sqrt[4]{n} < \sqrt{n} + \sqrt{n} + \sqrt{n} = 3\sqrt{n}$$

holds for any integer $n > 1$. Since $n \geq 3\sqrt{n}$ holds for any $n \geq 9$, the third inequality follows. ■

Remark. One may also use the AM-GM inequality to obtain the first inequality.

Example 1.41 (India RMO 2008 P3). Suppose a and b are real numbers such that the roots of the cubic equation $ax^3 - x^2 + bx - 1$ are positive real numbers. Prove that

- (i) $0 < 3ab \leq 1$,
- (ii) $b \geq \sqrt{3}$.

Solution 37. Let α, β, γ denote the roots of the polynomial $ax^3 - x^2 + bx - 1$. Note that

$$\alpha + \beta + \gamma = \frac{1}{a},$$

$$\begin{aligned}\alpha\beta + \beta\gamma + \gamma\alpha &= \frac{b}{a}, \\ \alpha\beta\gamma &= \frac{1}{a}\end{aligned}$$

holds. Since α, β, γ are positive, it follows that a, b are positive. Using

$$(\alpha + \beta + \gamma)^2 \geq 3(\alpha\beta + \beta\gamma + \gamma\alpha),$$

we obtain $\frac{1}{a^2} \geq \frac{3b}{a}$. Since a is positive, it follows that $3ab \leq 1$. Note that

$$\begin{aligned}(\alpha\beta + \beta\gamma + \gamma\alpha)^2 &= \alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 + 2\alpha\beta\gamma(\alpha + \beta + \gamma) \\ &\geq \frac{(\alpha\beta + \beta\gamma + \gamma\alpha)^2}{3} + 2\alpha\beta\gamma(\alpha + \beta + \gamma)\end{aligned}$$

holds, we get $\frac{b^2}{a^2} \geq \frac{b^2}{3a^2} + 2\frac{1}{a^2}$, which gives $b^2 \geq 3$. Since b is positive, we conclude $b \geq \sqrt{3}$. ■

Example 1.42 (India RMO 2011b P6). Find the largest real constant λ such that

$$\frac{\lambda abc}{a + b + c} \leq (a + b)^2 + (a + b + 4c)^2 \quad (2)$$

for all positive real numbers a, b, c .

Solution 38. Let λ be a nonnegative real number. Then the above inequality holds for all $a, b, c > 0$ if and only if

$$\lambda x^2 c \leq (2x + c)(4x^2 + 4(x + 2c)^2)$$

holds for all $x, c > 0$. Noting that both the sides are homogeneous of degree three in x and y , it follows that the above inequality holds if and only if

$$\lambda x^2 \leq (2x + 1)(4x^2 + 4(x + 2)^2)$$

holds for all $x > 0$. Observe that

$$\begin{aligned}(2x + 1)(4x^2 + 4(x + 2)^2) &= 8(2x + 1)(x^2 + 2x + 2) \\ &= 8(2x^3 + 5x^2 + 6x + 2).\end{aligned}$$

This shows that for $x > 0$, the above inequality is equivalent to

$$\frac{1}{2} \left(\frac{\lambda}{8} - 5 \right) \leq x + \frac{3}{x} + \frac{1}{x^2}.$$

Let $a, b > 0$ be such that

$$x + \frac{3}{x} + \frac{1}{x^2} = ax + \frac{3}{x} + 2bx + \frac{1}{x^2}$$

holds for any $x > 0$, and $(3/a)^3 = (1/b)^2$ holds, or equivalently, a, b satisfy $a + 2b = 1$ and $(3/a)^3 = (1/b)^2$. Note that this holds for $a = 3/4, b = 1/8$. Observe that

$$x + \frac{3}{x} + \frac{1}{x^2} = \frac{3}{4}x + \frac{3}{x} + \frac{1}{8}x + \frac{1}{8}x + \frac{1}{x^2} \geq 3 + \frac{3}{4} = \frac{15}{4},$$

where equality holds if and only if $x = 2$. This proves that the largest real constant λ satisfying the given inequality for all $a, b, c > 0$ satisfies

$$\frac{1}{2} \left(\frac{\lambda}{8} - 5 \right) = \frac{15}{4},$$

or equivalently, $\lambda = 100$ holds. ■

Example 1.43 (India RMO 2012e P4). Let a, b, c be positive real numbers such that $abc(a + b + c) = 3$. Prove that we have

$$(a + b)(b + c)(c + a) \geq 8.$$

Also determine the case of equality.

Solution 39. Note that

$$\begin{aligned} (a + b)(b + c)(c + a) &= (a + b + c)(ab + bc + ca) - abc \\ &= abc(a + b + c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) - abc \\ &= 3 \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) - abc \\ &\geq 9 \frac{1}{\sqrt[3]{abc}} - abc. \end{aligned}$$

Also note that

$$\begin{aligned} 3 &= abc(a + b + c) \\ &\geq 3(abc)^{4/3}, \end{aligned}$$

which gives $abc \leq 1$. Using the above inequalities, we obtain

$$\begin{aligned} (a + b)(b + c)(c + a) &\geq 9 - 1 \\ &= 8. \end{aligned}$$

This proves that

$$(a + b)(b + c)(c + a) \geq 8,$$

where equality holds if and only if all the prior inequalities are equalities, or equivalently, $a = b = c$. Using $abc(a + b + c) = 3$, it follows that a, b, c are equal if and only if they are equal to 1. This shows that $(a + b)(b + c)(c + a) = 8$ if and only if a, b, c are equal to 1. ■

Example 1.44 (India RMO 2012f P8). Let x, y, z be positive real numbers such that $2(xy + yz + zx) = xyz$. Prove that

$$\frac{1}{(x-2)(y-2)(z-2)} + \frac{8}{(x+2)(y+2)(z+2)} \leq \frac{1}{32}.$$

If a, b, c are positive real numbers such that $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$, then show that

$$\frac{1}{(a-1)(b-1)(c-1)} + \frac{8}{(a+1)(b+1)(c+1)} \leq \frac{1}{4}.$$

Solution 40. Put

$$x = 2a, y = 2b, z = 2c.$$

Note that a, b, c are positive real numbers and they satisfy $ab + bc + ca = abc$. Observe that

$$\begin{aligned} (a-1)(b-1)(c-1) &= abc - ab - bc - ca + a + b + c - 1 \\ &= a + b + c - 1, \\ (a+1)(b+1)(c+1) &= abc + ab + bc + ca + a + b + c + 1 \\ &= 2abc + a + b + c + 1. \end{aligned}$$

Using the AM-GM-HM inequality, we obtain

$$\frac{a+b+c}{3} \geq (abc)^{\frac{1}{3}} \geq \frac{3}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}},$$

which yields $a + b + c \geq 9$ and $abc \geq 3^3$. This implies that

$$\begin{aligned} &\frac{1}{(a-1)(b-1)(c-1)} + \frac{8}{(a+1)(b+1)(c+1)} \\ &\leq \frac{1}{9-1} + \frac{8}{2 \cdot 27 + 9 + 1} \\ &= \frac{1}{8} + \frac{8}{64} \\ &= \frac{1}{4}. \end{aligned}$$

■

Exercise 1.45 (ELMO 2013 P2). Let a, b, c be positive reals satisfying $a + b + c = \sqrt[3]{a} + \sqrt[3]{b} + \sqrt[3]{c}$. Prove that $a^a b^b c^c \geq 1$.

Walkthrough —**(a)****Solution 41.** Using the weighted AM-GM inequality, we obtain

$$1 = \sum_{\text{cyc}} \frac{a}{a+b+c} \cdot a^{-\frac{6}{7}} \geq (a^a b^b c^c)^{-\frac{6/7}{a+b+c}},$$

which yields $a^a b^b c^c \geq 1$. ■**Example 1.46 (India RMO 2014e P5).** Let a, b, c be positive real numbers such that

$$\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c} \leq 1. \quad (3)$$

Prove that

$$(1+a^2)(1+b^2)(1+c^2) \geq 125.$$

When does the equality hold?

Solution 42. The given inequality implies

$$\frac{a}{1+a} \geq \frac{1}{1+b} + \frac{1}{1+c} \geq \frac{2}{\sqrt{(1+b)(1+c)}},$$

where the second inequality holds if and only if $b = c$. Similar lower bounds for $\frac{b}{1+b}, \frac{c}{1+c}$ can be obtained, and multiplying them yields $abc \geq 8$, where equality holds if and only if $a = b = c = 2$. Note that

$$1 + a^2 = 1 + \frac{a^2}{4} + \frac{a^2}{4} + \frac{a^2}{4} + \frac{a^2}{4} \geq 5 \left(\frac{a^8}{2^8} \right)^{\frac{1}{5}}$$

holds, where equality occurs if and only if $a = 2$. We can find similar lower bounds for $1 + b^2, 1 + c^2$. Multiplying them, we get

$$(1+a^2)(1+b^2)(1+c^2) \geq 5^3 \left(\frac{a^8}{2^8} \frac{b^8}{2^8} \frac{c^8}{2^8} \right)^{\frac{1}{5}} = 5^3 \left(\frac{abc}{8} \right)^{\frac{8}{5}} \geq 125,$$

where equality occurs if and only if $a = b = c = 2$. ■**Example 1.47 (India RMO 2016c P2).** Let a, b, c be three distinct positive real numbers such that $abc = 1$. Prove that

$$\frac{a^3}{(a-b)(a-c)} + \frac{b^3}{(b-c)(b-a)} + \frac{c^3}{(c-a)(c-b)} \geq 3.$$

Solution 43. Note that

$$\begin{aligned}
 & \frac{a^3}{(a-b)(a-c)} + \frac{b^3}{(b-c)(b-a)} + \frac{c^3}{(c-a)(c-b)} \\
 &= -\frac{a^3(b-c) + b^3(c-a) + c^3(a-b)}{(a-b)(b-c)(c-a)} \\
 &= -\frac{a^3(b-c) + b^3(c-a) - c^3(b-c) - c^3(c-a)}{(a-b)(b-c)(c-a)} \\
 &= -\frac{(b^3 - c^3)(c-a) + (a^3 - c^3)(b-c)}{(a-b)(b-c)(c-a)} \\
 &= -\frac{b^2 + bc + c^2 - c^2 - a^2 - ca}{a-b} \\
 &= -\frac{b^2 + bc - a^2 - ca}{a-b} \\
 &= a + b + c \\
 &\geq 3\sqrt[3]{abc} \\
 &= 3.
 \end{aligned}$$

■

Remark. The above argument leads to following somewhat simpler solution. Observing that

$$\begin{aligned}
 \frac{1}{(c-a)(c-b)} &= \frac{(c-a) - (c-b)}{(b-a)(c-a)(c-b)} \\
 &= \frac{1}{(b-a)(c-b)} - \frac{1}{(b-a)(c-a)},
 \end{aligned}$$

we obtain

$$\begin{aligned}
 & \frac{a^3}{(a-b)(a-c)} + \frac{b^3}{(b-c)(b-a)} + \frac{c^3}{(c-a)(c-b)} \\
 &= \frac{a^3 - c^3}{(a-b)(a-c)} + \frac{b^3 - c^3}{(b-c)(b-a)} \\
 &= \frac{a^2 + ac + c^2}{a-b} - \frac{b^2 + bc + c^2}{a-b} \\
 &= a + b + c \\
 &\geq 3.
 \end{aligned}$$

Example 1.48 (India RMO 2016e P4). Let a, b, c be positive real numbers such that $a + b + c = 3$. Determine, with certainty, the largest possible value of the expression

$$\frac{a}{a^3 + b^2 + c} + \frac{b}{b^3 + c^2 + a} + \frac{c}{c^3 + a^2 + b}.$$

Solution 44. Note that

$$\begin{aligned}
 & \frac{a}{a^3 + b^2 + c} + \frac{b}{b^3 + c^2 + a} + \frac{c}{c^3 + a^2 + b} \\
 &= \frac{1}{a^2 + \frac{b^2}{a} + \frac{c}{a}} + \frac{1}{b^2 + \frac{c^2}{b} + \frac{a}{b}} + \frac{1}{c^2 + \frac{a^2}{c} + \frac{b}{c}} \\
 &\leq \frac{1 + a + ca}{(a + b + c)^2} + \frac{1 + b + ab}{(a + b + c)^2} + \frac{1 + c + bc}{(a + b + c)^2} \\
 &\quad (\text{by the Cauchy-Schwarz inequality}) \\
 &= \frac{3 + a + b + c + ab + bc + ca}{(a + b + c)^2} \\
 &\leq \frac{6 + \frac{(a+b+c)^2}{3}}{9} \\
 &= 1.
 \end{aligned}$$

Also note that if $a = b = c = 1$, then $a + b + c = 3$ and

$$\frac{a}{a^3 + b^2 + c} + \frac{b}{b^3 + c^2 + a} + \frac{c}{c^3 + a^2 + b} = 1.$$

This shows that the largest possible value of the given expression is equal to 1. ■

Example 1.49 (India RMO 2016f P5). Let x, y, z be non-negative real numbers such that $xyz = 1$. Prove that

$$(x^3 + 2y)(y^3 + 2z)(z^3 + 2x) \geq 27.$$

Solution 45. Applying the AM-GM inequality, we obtain

$$\begin{aligned}
 & (x^3 + 2y)(y^3 + 2z)(z^3 + 2x) \\
 &\geq (3\sqrt[3]{x^3y^2})(3\sqrt[3]{y^3z^2})(3\sqrt[3]{z^3x^2}) \\
 &= 27.
 \end{aligned}$$
■

Example 1.50 (India RMO 2019a P3). Let a, b, c be positive real numbers such that $a + b + c = 1$. Prove that

$$\frac{a}{a^2 + b^3 + c^3} + \frac{b}{b^2 + a^3 + c^3} + \frac{c}{c^2 + a^3 + b^3} \leq \frac{1}{5abc}.$$

Walkthrough — **Homogenize** the denominator and apply the AM-GM inequality.

Solution 46. Note that

$$\begin{aligned}
 \frac{a}{a^2 + b^3 + c^3} &= \frac{a}{a^2(a + b + c) + b^3 + c^3} \\
 &\leq \frac{a}{5\sqrt[5]{a^6 \cdot abc \cdot b^3 \cdot c^3}} \\
 &= \frac{1}{5\sqrt[5]{a^2 b^4 c^4}} \\
 &= \frac{1}{5abc} (a^3 bc)^{1/5} \\
 &\leq \frac{3a + b + c}{25abc}
 \end{aligned}$$

holds. Similarly, it follows that

$$\begin{aligned}
 \frac{b}{b^2 + a^3 + c^3} &\leq \frac{3b + a + c}{25abc}, \\
 \frac{c}{c^2 + a^3 + b^3} &\leq \frac{3c + a + b}{25abc}.
 \end{aligned}$$

Adding the above inequalities, we obtain

$$\begin{aligned}
 &\frac{a}{a^2 + b^3 + c^3} + \frac{b}{b^2 + a^3 + c^3} + \frac{c}{c^2 + a^3 + b^3} \\
 &\leq \frac{3a + b + c}{25abc} + \frac{3b + a + c}{25abc} + \frac{3c + a + b}{25abc} \\
 &= \frac{a + b + c}{5abc} \\
 &= \frac{1}{5abc}.
 \end{aligned}$$

■

Exercise 1.51 (Canadian Mathematical Olympiad 2021 P2). Let $n \geq 2$ be some fixed positive integer and suppose that $a_1, a_2, \dots, a_n$ are positive real numbers satisfying

$$a_1 + a_2 + \dots + a_n = 2^n - 1.$$

Find the minimum possible value of

$$\frac{a_1}{1} + \frac{a_2}{1 + a_1} + \frac{a_3}{1 + a_1 + a_2} + \dots + \frac{a_n}{1 + a_1 + a_2 + \dots + a_{n-1}}.$$

Walkthrough —**(a)****Solution 47.** Note that

$$\begin{aligned}
& \frac{a_1}{1} + \frac{a_2}{1+a_1} + \frac{a_3}{1+a_1+a_2} + \cdots + \frac{a_n}{1+a_1+a_2+\cdots+a_{n-1}} \\
&= \frac{1+a_1}{1} + \frac{1+a_1+a_2}{1+a_1} + \frac{1+a_1+a_2+a_3}{1+a_1+a_2} \\
&\quad + \cdots + \frac{1+a_1+a_2+\cdots+a_n}{1+a_1+a_2+\cdots+a_{n-1}} - n \\
&\geq n \sqrt[n]{1+a_1+a_2+\cdots+a_n} - n \\
&= n \sqrt[n]{2^n} - n \\
&= n.
\end{aligned}$$

This shows that the minimum possible value of the given expression is at least n .

Also note that if $a_i = 2^{i-1}$ for all $1 \leq i \leq n$, then $a_1 + a_2 + \cdots + a_n = 2^n - 1$ holds, and

$$\begin{aligned}
& \frac{a_1}{1} + \frac{a_2}{1+a_1} + \frac{a_3}{1+a_1+a_2} + \cdots + \frac{a_n}{1+a_1+a_2+\cdots+a_{n-1}} \\
&= \frac{1+a_1}{1} + \frac{1+a_1+a_2}{1+a_1} + \frac{1+a_1+a_2+a_3}{1+a_1+a_2} \\
&\quad + \cdots + \frac{1+a_1+a_2+\cdots+a_n}{1+a_1+a_2+\cdots+a_{n-1}} - n \\
&= 2n - n \\
&= n.
\end{aligned}$$

This proves that the minimum possible value of the given expression is equal to n . ■

§1.7 Bunching terms**Example 1.52.** Show that

$$\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{2^m} \geq \frac{m}{2} \quad \text{for } m \geq 1.$$

Use it to show that $\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n}$ becomes arbitrarily large as n increases indefinitely.

Walkthrough — Observe that

$$\begin{aligned}
 & \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{2^m} \\
 &= \frac{1}{2} \\
 &+ \left(\frac{1}{3} + \frac{1}{4} \right) \\
 &+ \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} \right) \\
 &+ \cdots \\
 &+ \left(\frac{1}{2^{m-1}} + \frac{1}{2^{m-1}+1} + \cdots + \frac{1}{2^m} \right).
 \end{aligned}$$

Show that the sum in each parenthesis is at most $\frac{1}{2}$.

Example 1.53 (India RMO 1992 P7). Show that

$$1 < \frac{1}{1001} + \frac{1}{1002} + \cdots + \frac{1}{3001} < \frac{4}{3}$$

Solution 48. To obtain the lower bound, note that

$$\begin{aligned}
 & \frac{1}{1001} + \frac{1}{1002} + \cdots + \frac{1}{3001} \\
 &> \frac{1}{1001} + \frac{1}{1002} + \cdots + \frac{1}{3000} \\
 &= \left(\frac{1}{1001} + \cdots + \frac{1}{1250} \right) + \left(\frac{1}{1251} + \cdots + \frac{1}{1500} \right) + \left(\frac{1}{1501} + \cdots + \frac{1}{1750} \right) \\
 &+ \left(\frac{1}{1751} + \cdots + \frac{1}{2000} \right) + \left(\frac{1}{2001} + \cdots + \frac{1}{2250} \right) + \left(\frac{1}{2251} + \cdots + \frac{1}{2500} \right) \\
 &+ \left(\frac{1}{2501} + \cdots + \frac{1}{2750} \right) + \left(\frac{1}{2751} + \cdots + \frac{1}{3000} \right) \\
 &> \frac{250}{1250} + \frac{250}{1500} + \frac{250}{1750} + \frac{250}{2000} + \frac{250}{2250} + \frac{250}{2500} + \frac{250}{2750} + \frac{250}{3000} \\
 &> \frac{1}{5} + \frac{1}{6} + \cdots + \frac{1}{12} = 17 \left(\frac{1}{60} + \frac{1}{66} + \frac{1}{70} + \frac{1}{72} \right) > 1.
 \end{aligned}$$

To get the upper bound, note that

$$\begin{aligned}
 & \frac{1}{1001} + \frac{1}{1002} + \cdots + \frac{1}{3001} \\
 &< \frac{1}{1000} + \frac{1}{1001} + \cdots + \frac{1}{3000} \\
 &< \left(\frac{1}{1000} + \frac{1}{1001} + \cdots + \frac{1}{1499} \right) + \left(\frac{1}{1500} + \frac{1}{1501} + \cdots + \frac{1}{1999} \right)
 \end{aligned}$$

$$\begin{aligned}
& + \left(\frac{1}{2000} + \frac{1}{2001} + \cdots + \frac{1}{2499} \right) + \left(\frac{1}{2500} + \frac{1}{2501} + \cdots + \frac{1}{2999} \right) + \frac{1}{3000} \\
& < \frac{500}{1500} + \frac{500}{2000} + \frac{500}{2500} + \frac{500}{3000} + \frac{1}{3000} \\
& = \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{3000} \\
& = \frac{57}{60} + \frac{1}{3000} \\
& < 1 \\
& < \frac{4}{3}.
\end{aligned}$$

■

Example 1.54 (Canada CMO 1998 P3, India RMO 1998 P3). Prove the following inequality for every natural number $n \geq 2$:

$$\frac{1}{n+1} \left(1 + \frac{1}{3} + \frac{1}{5} + \cdots + \frac{1}{2n-1} \right) > \frac{1}{n} \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \cdots + \frac{1}{2n} \right).$$

Solution 49. Let us establish the following Claim.

Claim — For any $1 \leq k \leq n$, the inequality

$$\frac{1}{n+1} \left(\frac{1}{2k-1} + \frac{1}{2n-2k+1} \right) \geq \frac{1}{n} \left(\frac{1}{2k} + \frac{1}{2n-2k+2} \right)$$

holds. Moreover, this inequality is strict if $k = 1$ and $n \geq 2$.

Proof of the Claim. Clearing the denominators, it follows that the above inequality is equivalent to

$$4n^2k(n-k+1) \geq (n+1)^2(2k-1)(2n-2k+1).$$

Note that

$$\begin{aligned}
& 4n^2k(n-k+1) - (n+1)^2(2k-1)(2n-2k+1) \\
& = n^2(4nk - 4k^2 + 4k - (4nk - 4k^2 + 4k - 2n - 1)) \\
& \quad - (2n+1)(2k-1)(2n-2k+1) \\
& = n^2(2n+1) - (2n+1)(2k-1)(2n-2k+1) \\
& = (2n+1)(n^2 - (4nk - 4k^2 + 4k - 2n - 1)) \\
& = (2n+1)((n-2k)^2 + 2(n-2k) + 1) \\
& = (2n+1)(n-2k+1)^2,
\end{aligned}$$

which is nonnegative for any $1 \leq k \leq n$. This proves the inequality. Moreover, if $k = 1$ and $n \geq 2$, then

$$4n^2k(n - k + 1) - (n + 1)^2(2k - 1)(2n - 2k + 1) > 0,$$

which proves the second statement. □

The Claim implies the given inequality. ■

§1.8 Look at the difference

Example 1.55. Show that if $a > b > 0$, then

$$a + \frac{1}{(a - b)b} \geq 3.$$

Solution 50. Applying the AM-GM inequality, we obtain

$$a + \frac{1}{(a - b)b} = (a - b) + b + \frac{1}{(a - b)b} \geq 3.$$

Example 1.56. Prove that if $a < b < c < d$, then $(a + b + c + d)^2 > 8(ac + bd)$.

Solution 51. Let x, y, z denote the real numbers satisfying

$$\begin{aligned} b &= a + x, \\ c &= b + y, \\ d &= c + z. \end{aligned}$$

Note that

$$\begin{aligned} a + b + c + d &= a + b + 2c + (d - c) \\ &= a + 3b + 2(c - b) + (d - c) \\ &= 4a + 3(b - a) + 2(c - b) + (d - c) \\ &= 4a + 3x + 2y + z, \\ ac + bd &= ac + (a + x)(c + z) \\ &= 2ac + cx + az + zx \\ &= 2a(b + y) + x(b + y) + az + zx \\ &= 2a(a + x + y) + x(a + x + y) + az + zx \\ &= 2a^2 + 2ax + 2ay + ax + x^2 + xy + az + zx \\ &= 2a^2 + 3ax + 2ay + zx + x^2 + xy + az \end{aligned}$$

hold. This shows that

$$\begin{aligned}
 & (a + b + c + d)^2 - 8(ac + bd) \\
 &= (4a + 3x + 2y + z)^2 - 8(2a^2 + 3ax + 2ay + zx + x^2 + xy + az) \\
 &= 9x^2 + 4y^2 + z^2 + 24ax + 16ay + 8az + 12xy + 4yz + 6zx \\
 &\quad - 24ax - 16ay - 8zx - 8x^2 - 8xy - 8az \\
 &= x^2 + 4y^2 + z^2 + 4xy + 4yz - 2zx \\
 &= (x - z)^2 + 4y(x + y + z) \\
 &> 0.
 \end{aligned}$$

■

Example 1.57 (India RMO 2004 P7). Let x and y be positive real numbers such that $y^3 + y \leq x - x^3$. Prove that

1. $y < x < 1$ and
2. $x^2 + y^2 < 1$.

Solution 52. Note that $x - y \geq x^3 + y^3 > 0$ holds, which shows that $x > y$. Also note that $x - x^3 \geq y^3 + y > 0$ holds, which implies that $x(1 - x^2) > 0$. Since x is positive, we obtain $x < 1$. This gives $0 < y < x < 1$.

Write $x = y + t$ with $t > 0$. The inequality $y^3 + y \leq x - x^3$ yields

$$2y^3 + 3y^2t + 3yt + t^3 \leq t.$$

Note that the inequality $x^2 + y^2 < 1$ is equivalent to $2y^2 + 2yt + t^2 < 1$. Since $y + t$ is positive, it is equivalent to $(y + t)(2y^2 + 2yt + t^2) < y + t$. Observe that

$$\begin{aligned}
 (y + t)(2y^2 + 2yt + t^2) &= 2y^3 + 4y^2t + 3yt^2 + t^3 \\
 &\leq y^2t + 3yt^2 + t - 3yt \\
 &= y^2t - 3yt(1 - t) + t \\
 &< y + t \quad (\text{using } 0 < y, t < 1).
 \end{aligned}$$

This completes the proof. ■

Example 1.58 (India RMO 2017b P5). If $a, b, c, d \in \mathbb{R}$ such that $a > b > c > d > 0$ and $a + d = b + c$; then prove that

$$\frac{(a + b) - (c + d)}{\sqrt{2}} > \sqrt{a^2 + b^2} - \sqrt{c^2 + d^2}.$$

Solution 53. Put $m = a + d$. Using $a + d = b + c$ and $a > b > c > d > 0$, it follows that for some real numbers $0 < y < x < m$,

$$a = m + x, b = m + y, c = m - y, d = m - x$$

hold. Note that the given inequality reduces to

$$\sqrt{2}(x + y) > \sqrt{(m + x)^2 + (m + y)^2} - \sqrt{(m - x)^2 + (m - y)^2}.$$

Observe that

$$\begin{aligned} & \sqrt{(m + x)^2 + (m + y)^2} - \sqrt{(m - x)^2 + (m - y)^2} \\ &= \frac{(m + x)^2 + (m + y)^2 - (m - x)^2 - (m - y)^2}{\sqrt{(m + x)^2 + (m + y)^2} + \sqrt{(m - x)^2 + (m - y)^2}} \\ &= \frac{4m(x + y)}{\sqrt{(m + x)^2 + (m + y)^2} + \sqrt{(m - x)^2 + (m - y)^2}}. \end{aligned}$$

Hence, it suffices to prove that

$$\sqrt{(m + x)^2 + (m + y)^2} + \sqrt{(m - x)^2 + (m - y)^2} > 2\sqrt{2}m.$$

Note that

$$\begin{aligned} & \sqrt{(m + x)^2 + (m + y)^2} + \sqrt{(m - x)^2 + (m - y)^2} \\ & > \frac{m + x + m + y}{\sqrt{2}} + \frac{m - x + m - y}{\sqrt{2}} \\ &= 2\sqrt{2}m \end{aligned}$$

hold. This completes the proof. ■

Example 1.59 (India RMO 2024b P4). Let a_1, a_2, a_3, a_4 be real numbers such that $a_1^2 + a_2^2 + a_3^2 + a_4^2 = 1$. Show that there exist i, j with $1 \leq i < j \leq 4$, such that $(a_i - a_j)^2 \leq \frac{1}{5}$.

Solution 54. Reordering a_1, a_2, a_3, a_4 if necessary, we assume that $a_1 \leq a_2 \leq a_3 \leq a_4$ holds. Put

$$a_2 = a_1 + x,$$

$$a_3 = a_2 + y,$$

$$a_4 = a_3 + z.$$

Note that

$$a_2^2 = a_1^2 + x^2 + 2a_1x,$$

$$a_3^2 = (a_1 + x + y)^2$$

$$\begin{aligned}
&= a_1^2 + x^2 + y^2 + 2a_1x + 2a_1y + 2xy, \\
a_4^2 &= (a_1 + x + y + z)^2 \\
&= a_1^2 + x^2 + y^2 + z^2 + 2a_1x + 2a_1y + 2a_1z + 2xy + 2yz + 2zx
\end{aligned}$$

hold. This gives

$$\begin{aligned}
&a_1^2 + a_2^2 + a_3^2 + a_4^2 \\
&= a_1^2 \\
&\quad + a_1^2 + x^2 + 2a_1x \\
&\quad + a_1^2 + x^2 + y^2 + 2a_1x + 2a_1y + 2xy \\
&\quad + a_1^2 + x^2 + y^2 + z^2 + 2a_1x + 2a_1y + 2a_1z + 2xy + 2yz + 2zx \\
&= 4a_1^2 + 3x^2 + 2y^2 + z^2 + 2a_1(3x + 2y + z) + 4xy + 2yz + 2zx \\
&= \left(2a_1 + \frac{3x + 2y + z}{2}\right)^2 - \frac{(3x + 2y + z)^2}{4} \\
&\quad + 3x^2 + 2y^2 + z^2 + 4xy + 2yz + 2zx \\
&= \left(2a_1 + \frac{3x + 2y + z}{2}\right)^2 - \left(\frac{9}{4}x^2 + y^2 + \frac{1}{4}z^2 + 3xy + yz + \frac{3}{2}zx\right) \\
&\quad + 3x^2 + 2y^2 + z^2 + 4xy + 2yz + 2zx \\
&= \left(2a_1 + \frac{3x + 2y + z}{2}\right)^2 + \frac{3}{4}x^2 + y^2 + \frac{3}{4}z^2 + xy + yz + \frac{1}{2}zx.
\end{aligned}$$

If $x > \frac{1}{\sqrt{5}}, y > \frac{1}{\sqrt{5}}, z > \frac{1}{\sqrt{5}}$ holds, then we would obtain

$$\begin{aligned}
1 &> \left(2a_1 + \frac{3x + 2y + z}{2}\right)^2 + \left(\frac{3}{4} + 1 + \frac{3}{4} + 1 + 1 + \frac{1}{2}\right) \frac{1}{5} \\
&= \left(2a_1 + \frac{3x + 2y + z}{2}\right)^2 + 1 \\
&\geq 1,
\end{aligned}$$

which yields $1 > 1$, which is impossible. This shows that at least one of x, y, z is less than or equal to $\frac{1}{\sqrt{5}}$. ■

§1.9 Algebraic substitutions

Example 1.60 (India RMO 2016a P2, cf. India RMO 2016b P2). Let a, b, c be positive real numbers such that

$$\frac{a}{1+a} + \frac{b}{1+b} + \frac{c}{1+c} = 1.$$

Prove that $abc \leq \frac{1}{8}$.

Example 1.61 (India RMO 2016b P2, cf. India RMO 2016a P2). Let a, b, c be positive real numbers such that

$$\frac{a}{1+a} + \frac{b}{1+b} + \frac{c}{1+c} = 1.$$

Prove that $abc \leq \frac{1}{8}$.

Solution 55. Put

$$x = \frac{a}{1+a}, y = \frac{b}{1+b}, z = \frac{c}{1+c}.$$

It follows that x, y, z are positive real numbers satisfying $x + y + z = 1$. Note that

$$a = \frac{x}{1-x}, b = \frac{y}{1-y}, c = \frac{z}{1-z}.$$

We need to show that $8abc \leq 1$, which is equivalent to $8xyz \leq (1-x)(1-y)(1-z)$. Using $x + y + z = 1$, it follows that the above inequality is equivalent to

$$8xyz \leq (x+y)(y+z)(z+x),$$

which follows from the AM-GM inequality. ■

Remark. A careful reading of the above argument shows that the substitution

$$\frac{x}{x+y+z} = \frac{a}{1+a}, \frac{y}{x+y+z} = \frac{b}{1+b}, \frac{z}{x+y+z} = \frac{c}{1+c},$$

or equivalently, the substitution

$$a = \frac{x}{y+z}, b = \frac{y}{z+x}, c = \frac{z}{x+y}$$

reduces $8abc \leq 1$ to the inequality $8xyz \leq (x+y)(y+z)(z+x)$.

We refer to Example 1.27.

§1.10 Ravi substitution

In Fig. 1, the incircle touches the sides of the triangle ABC at the points D, E, F . Note that $AF = AE, BF = BD, CD = CE$ holds, and let us denote them by x, y, z respectively. This gives

$$a = BC = y + z, b = CA = z + x, c = AB = x + y$$

with $x, y, z > 0$.

Example 1.62 (India RMO 1996 P5). Let ABC be a triangle and h_a the altitude through A . Prove that

$$(b+c)^2 \geq a^2 + 4h_a^2.$$

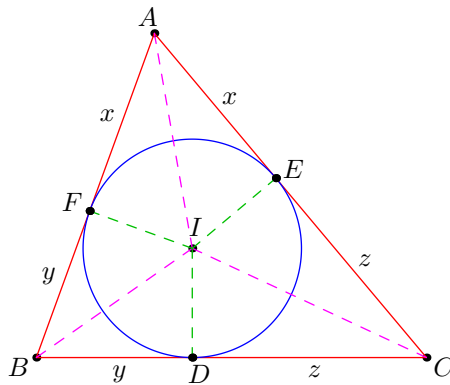


Figure 1: Ravi substitution

Solution 56. Since $\Delta = \frac{1}{2}ah_a$, the inequality reduces to

$$(b+c)^2 - a^2 \geq \frac{1}{a^2} 16\Delta^2,$$

which is equivalent to

$$a^2 \geq (c+a-b)(a+b-c).$$

This follows from the AM-GM inequality. ■

Remark. To prove the inequality $a^2 \geq (c+a-b)(a+b-c)$, one could substitute

$$a = y + z, b = z + x, c = x + y,$$

and note that the above inequality is equivalent to

$$(y+z)^2 \geq 4yz,$$

which holds since $(y-z)^2 \geq 0$.

Example 1.63 (India RMO 1999 P5). If a, b, c are the sides of a triangle, prove the following inequality:

$$\frac{a}{c+a-b} + \frac{b}{a+b-c} + \frac{c}{b+c-a} \geq 3.$$

Solution 57. Let us use Ravi substitution, that is, let x, y, z be real numbers satisfying

$$a = y + z, b = z + x, c = x + y.$$

Note that x, y, z are positive by the triangle inequality. Observe that

$$\begin{aligned} & \frac{a}{c+a-b} + \frac{b}{a+b-c} + \frac{c}{b+c-a} \\ &= \frac{y+z}{2y} + \frac{z+x}{2z} + \frac{x+y}{2x} \\ &= \frac{3}{2} + \frac{1}{2} \left(\frac{y}{x} + \frac{z}{y} + \frac{x}{z} \right) \\ &\geq \frac{3}{2} + \frac{3}{2} \quad (\text{by the AM-GM inequality}) \\ &= 3. \end{aligned}$$

■

Example 1.64 (India RMO 2001 P6). If x, y, z are the sides of a triangle, then prove that

$$|x^2(y-z) + y^2(z-x) + z^2(x-y)| < xyz.$$

Solution 58. Note that

$$\begin{aligned} & x^2(y-z) + y^2(z-x) + z^2(x-y) \\ &= x^2y - zx^2 + y^2z - xy^2 + z^2x - yz^2 \\ &= (x^2 - z^2)y + y^2(z-x) + zx(z-x) \\ &= (z-x)(y^2 + zx - y(z+x)) \\ &= (z-x)(y-z)(y-x). \end{aligned}$$

Since

$$|x^2(y-z) + y^2(z-x) + z^2(x-y)|, xyz$$

are symmetric in x, y, z , without loss of generality, we may (and do) assume that $x \leq y \leq z$. Note that $|(z-x)(y-z)(y-x)| < xyz$ is immediate if any two of x, y, z are equal. It remains to consider the case that x, y, z are distinct, which we assume from now on. Let us use Ravi substitution, that is, let a, b, c be real numbers satisfying

$$x = b + c, y = c + a, z = a + b,$$

where a, b, c are positive by the triangle inequality. Using $x < y < z$, we obtain $a > b > c$. Note that

$$\begin{aligned} |(z-x)(y-z)(y-x)| &= (a-b)(b-c)(a-c) \\ &< aba \\ &< (a+c)(b+c)(a+b) \\ &= xyz, \end{aligned}$$

where the first inequality follows since $0 < a-b < a, 0 < b-c < b, 0 < a-c < c$ holds. ■

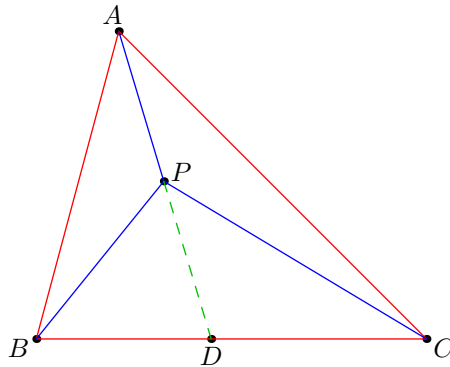


Figure 2: India RMO 1990 P5, Example 1.48

§1.11 Triangle inequality

Example 1.65. For real numbers x, y, z , show that

$$|x| + |y| + |z| \leq |x + y - z| + |y + z - x| + |z + x - y|.$$

Solution 59. Put

$$x = \frac{b+c}{2}, y = \frac{c+a}{2}, z = \frac{a+b}{2}.$$

In other words, let a, b, c be real numbers defined by

$$a = y + z - x, b = z + x - y, c = x + y - z.$$

Applying the triangle inequality, we obtain

$$|a| + |b| \geq 2|z|, |b| + |c| \geq 2|x|, |c| + |a| \geq 2|y|.$$

The result follows by adding them. ■

Example 1.66 (India RMO 1990 P5). Let P be a point inside a triangle ABC . Let s denote the semiperimeter $\frac{1}{2}(AB + BC + CA)$. Prove that

$$s < AP + BP + CP < 2s.$$

Solution 60. Applying the triangle inequality to PBC , we get $BC < BP + CP$. Similarly, the inequalities $CA < CP + AP$, $AB < AP + BP$ follow. Adding them yields

$$AB + BC + CA < 2(AP + BP + CP),$$

or equivalently, $s < AP + BP + CP$ holds. Let D denote the point on BC such that A, P, D are collinear. Note that

$$AP + BP < AP + BD + DP \quad (\text{applying the triangle inequality to } BDP)$$

$$\begin{aligned}
 &= AD + BD \\
 &< AC + DC + BD \quad (\text{applying the triangle inequality to } ADC) \\
 &= AC + BC
 \end{aligned}$$

hold. Similarly, it follows that

$$BP + CP < AB + CA, CP + AP < AB + BC.$$

Adding these inequalities, we obtain

$$AP + BP + CP < AB + BC + CA = 2s.$$

■

Example 1.67 (India RMO 1995 P5). Show that for any triangle ABC , the following inequality is true:

$$a^2 + b^2 + c^2 > \sqrt{3} \max\{|a^2 - b^2|, |b^2 - c^2|, |c^2 - a^2|\}.$$

Solution 61. Since the above inequality is symmetric with respect to a, b, c , without loss of generality, we assume that $a > b > c$. Thus we need to show that

$$a^2 + b^2 + c^2 > \sqrt{3}(a^2 - c^2)$$

holds. Using the triangle inequality, we obtain $b > a - c > 0$. This yields

$$\begin{aligned}
 a^2 + b^2 + c^2 - \sqrt{3}(a^2 - c^2) &> a^2 + (a - c)^2 + c^2 - \sqrt{3}(a^2 - c^2) \\
 &= 2(a^2 + c^2 - ca) - \sqrt{3}(a^2 - c^2) \\
 &= (2 - \sqrt{3})a^2 + (2 + \sqrt{3})c^2 - 2ca \\
 &= (2 - \sqrt{3})(a^2 - 2ac(2 + \sqrt{3}) + (2 + \sqrt{3})^2) \\
 &= (2 - \sqrt{3})(a - (2 + \sqrt{3})c)^2 \\
 &\geq 0.
 \end{aligned}$$

This completes the proof. ■

Example 1.68 (India RMO 1997 P5). Let x, y and z be three distinct real positive numbers. Determine with proof whether or not the three real numbers

$$\left| \frac{x}{y} - \frac{y}{x} \right|, \left| \frac{y}{z} - \frac{z}{y} \right|, \left| \frac{z}{x} - \frac{x}{z} \right|$$

can be the lengths of the sides of a triangle.

Solution 62. Without loss of generality, let us assume that $x > y > z$. Note that

$$\begin{aligned}
 & z(x^2 - y^2) + x(y^2 - z^2) - y(x^2 - z^2) \\
 &= zx(x - z) + (x - z)y^2 - y(x^2 - z^2) \\
 &= (x - z)(zx + y^2 - y(x + z)) \\
 &= (x - z)(y - x)(y - z) \\
 &< 0.
 \end{aligned}$$

By the triangle inequality, it follows that the given three numbers cannot be the lengths of the sides of a triangle. ■

Example 1.69 (India RMO 2009 P5). A convex polygon is such that the distance between any two vertices does not exceed 1.

- (i) Prove that the distance between any two points on the boundary of the polygon does not exceed 1.
- (ii) If X and Y are two distinct points inside the polygon, prove that there exists a point Z on the boundary of the polygon such that $XZ + YZ \leq 1$.

Solution 63. Note that for any four complex numbers z_1, z_2, z_3, z_4 and any $0 \leq t, s \leq 1$, we have

$$\begin{aligned}
 & (tz_1 + (1 - t)z_2) - (sz_3 + (1 - s)z_4) \\
 &= t(z_1 - z_3) + (t - s)z_3 + (1 - s)(z_2 - z_4) + (s - t)z_2 \\
 &= t(z_1 - z_3) + (s - t)(z_3 - z_2) + (1 - s)(z_2 - z_4)
 \end{aligned}$$

and

$$\begin{aligned}
 & (tz_1 + (1 - t)z_2) - (sz_3 + (1 - s)z_4) \\
 &= s(z_1 - z_3) + (t - s)z_1 + (1 - t)(z_2 - z_4) + (s - t)z_4 \\
 &= s(z_1 - z_3) + (t - s)(z_1 - z_4) + (1 - t)(z_2 - z_4).
 \end{aligned}$$

If the distance between no two of z_1, z_2, z_3, z_4 exceeds 1, then

$$|(tz_1 + (1 - t)z_2) - (sz_3 + (1 - s)z_4)| \leq \begin{cases} t + (s - t) + (1 - s) = 1 & \text{if } s \geq t, \\ s + (t - s) + (1 - t) = 1 & \text{if } t \geq s. \end{cases}$$

This proves part (i).

Extending XY to the boundary of the polygon, we find two points Z_1, Z_2 on the boundary such that Z_1, X, Y, Z_2 are collinear. Note that

$$(XZ_1 + YZ_1) + (XZ_2 + YZ_2) = (XZ_1 + XZ_2) + (YZ_1 + YZ_2)$$

$$\begin{aligned}
 &= 2Z_1Z_2 \\
 &\leq 2,
 \end{aligned}$$

which shows that $XZ_1 + YZ_1 \leq 1$ or $XZ_2 + YZ_2 \leq 1$ holds. This proves part (ii). ■

Example 1.70 (India RMO 2014e P1). Three positive real numbers a, b, c are such that $a^2 + 5b^2 + 4c^2 - 4ab - 4bc = 0$. Can a, b, c be the lengths of the sides of a triangle? Justify your answer.

Solution 64. The given condition is equivalent to

$$(a - 2b)^2 + (b - 2c)^2 = 0.$$

Since a, b, c are real numbers, this gives $b = 2c, a = 2b = 2c$. Note that $b + c = 3c < 2c = a$ holds. So by the triangle inequality, it follows that a, b, c cannot be the lengths of the sides of a triangle. ■

Exercise 1.71 (IOQM 2024 P10, AoPS). Determine the number of positive integral values of p for which there exists a triangle with sides a, b and c which satisfy

$$a^2 + (p^2 + 9)b^2 + 9c^2 - 6ab - 6pbc = 0$$

Summary — Complete squares and use the triangle inequality.

Walkthrough —

(a) Note that

$$a^2 + (p^2 + 9)b^2 + 9c^2 - 6ab - 6pbc = (a - 3b)^2 + (pb - 3c)^2.$$

(b) This gives $a = 3b$ and

$$c = \frac{p}{3}b.$$

(c) Use the inequality^a $a + b > c$, to obtain

$$3b + b > \frac{p}{3}b,$$

which yields $p \leq 11$.

(d) Use the inequality^b $b + c > a$, to get

$$b + \frac{p}{3}b > 3b,$$

implying $p \geq 7$.

(e) It reduces to finding the number of positive integers p such that $7 \leq p \leq 11$.

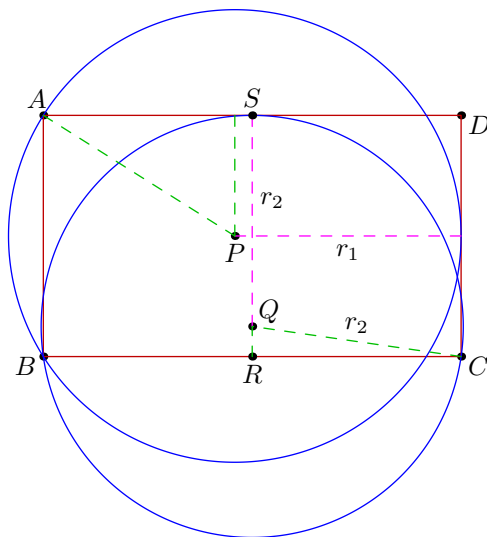


Figure 3: India RMO 1993 P4, Example 1.72

What about the condition $c + a > b$? Anything else to be taken care of?

^aHow does it follow?

^bHow does it follow?

Solution 65. ■

§1.12 Geometric inequalities

Example 1.72 (India RMO 1993 P4). Let $ABCD$ be a rectangle with $AB = a$ and $BC = b$. Suppose r_1 is the radius of the circle passing through A and B and touching CD ; and similarly r_2 is the radius of the circle passing through B and C and touching AD . Show that

$$r_1 + r_2 \geq \frac{5}{8}(a + b).$$

Solution 66. Since QS is perpendicular to AD , it is parallel to CD . Let SQ produced meet BC at R . Since RS is parallel to CD and $\angle RCD$ is a right-angle, it follows that QR is perpendicular to BC . This gives $QR^2 + RC^2 = QC^2$, which yields

$$(a - r_2)^2 + \left(\frac{b}{2}\right)^2 = r_2^2.$$

This implies that

$$r_2 = \frac{a}{2} + \frac{b^2}{8a}.$$

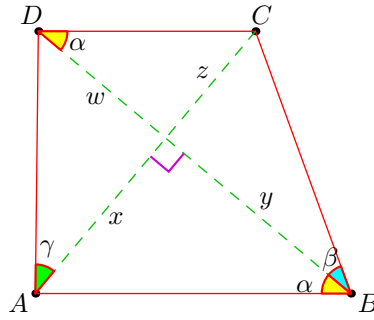


Figure 4: India RMO 1997 P4, Example 1.73

A similar argument shows that

$$r_1 = \frac{b}{2} + \frac{a^2}{8b}.$$

It follows that

$$\begin{aligned} r_1 + r_2 &= \frac{1}{2}(a + b) + \frac{1}{8} \left(\frac{a^2}{b} + \frac{b^2}{a} \right) \\ &\geq \frac{1}{2}(a + b) + \frac{1}{8} \frac{(a + b)^2}{a + b} \quad (\text{by the Cauchy-Schwarz inequality}) \\ &= \frac{5}{8}(a + b). \end{aligned}$$

This completed the proof. ■

Example 1.73 (India RMO 1997 P4). In a quadrilateral $ABCD$, it is given that AB is parallel to CD and the diagonals AC and BD are perpendicular to each other. Show that

- (a) $AD \cdot BC \geq AB \cdot CD$,
- (b) $AD + BC \geq AB + CD$.

Solution 67. Note that the inequality $AD \cdot BC \geq AB \cdot CD$ is equivalent to

$$(x^2 + w^2)(y^2 + z^2) \geq (x^2 + y^2)(z^2 + w^2),$$

which reduces to $(x^2 - z^2)(y^2 - w^2) \geq 0$. It suffices to show that $(x - z)(y - w) \geq 0$ holds. Since AB is parallel to CD , it follows that the triangles ABP and CDP are similar, and the lengths of their sides satisfy

$$\frac{x}{z} = \frac{y}{w}.$$

Consequently, the inequality $(x - z)(y - w) \geq 0$ holds.

To prove the second inequality, note that

$$\begin{aligned}(AD + BC)^2 &= AD^2 + BC^2 + 2AD \cdot BC \\ &= x^2 + w^2 + y^2 + z^2 + 2AD \cdot BC \\ &\geq x^2 + y^2 + z^2 + w^2 + 2AB \cdot CD \\ &= AB^2 + CD^2 + 2AB \cdot CD \\ &= (AB + CD)^2,\end{aligned}$$

which implies $AD + BC \geq AB + CD$. ■

Example 1.74 (India RMO 2013b P5). Let $n \geq 3$ be a natural number and let P be a polygon with n sides. Let $a_1, a_2, \dots, a_n$ be the lengths of sides of P and let p be its perimeter. Prove that

$$\frac{a_1}{p - a_1} + \frac{a_2}{p - a_2} + \dots + \frac{a_n}{p - a_n} < 2.$$

Solution 68. By a repeated application of the triangle inequality, it follows that $a_i < p - a_i$ for all $1 \leq i \leq n$. Note that for any positive real numbers a, b, c , the inequality

$$\frac{a}{b} < \frac{a + c}{b + c}$$

holds. It follows that

$$\frac{a_1}{p - a_1} + \frac{a_2}{p - a_2} + \dots + \frac{a_n}{p - a_n} < \frac{2a_1}{p} + \frac{2a_2}{p} + \dots + \frac{2a_n}{p} = 2$$

holds. ■

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